

emission intensity by 42% by 2030, from a 2005 baseline, and is aiming for carbon neutrality by 2050.

“These targets are aligned with the International Energy Agency’s (IEA) sustainable development scenario for the iron and steel sector for India,” Prabodha Acharya states. A major component of JSW’s decarbonisation strategy is based on the energy transition to renewable energy. “Central to our decarbonisation journey is our target of renewable electricity capacity of 10 GW by 2030,” says Prabodha Acharya. He adds that the company is also working to implement alternative fuels such as natural gas, biochar, and waste gases in its ironmaking processes to improve energy efficiency.

JSW Steel has integrated sustainable development into its business approach, focusing on contributing responsibly to society and the environment without compromising future generations. Jayant Acharya, Joint Managing Director and CEO of JSW Steel, explains, “Our sustainability vision is rooted in acting socially, ethically, and environmentally responsible, ensuring the needs of future generations aren’t compromised.” To implement this vision, JSW Steel has developed a sustainability framework covering 17 focus areas across environment, social, and governance (ESG) aspects. In terms of renewable energy, JSW Steel has operationalised around 800 megawatts (MW) of capacity, with plans to increase this to 2.5 gigawatts (GW).

As the company expands production, it aims to power all its operations through 10 GW of renewable capacity by 2030. It is working on green hydrogen steelmaking at its Vijayanagar plant in Karnataka, using Direct Reduced Iron (DRI) technology to lower emissions. “We are in the process of implementing green hydrogen-based steel making through

our DRI plant at our flagship unit Vijayanagar in the state of Karnataka,” Jayant Acharya says.

The company launched a specific programme called Sustainable Energy Environment and Decarbonisation (SEED) at operation sites to bring changes at the grassroots level. “This has resulted in CO₂ emission reduction of more than 3.5 million tonnes,” Jayant Acharya says. JSW is investing in modernising existing plants and building new ones with a focus on cleaner technologies.

steel, marking the transition to a second phase where JSW will explore emerging technologies for deeper emission cuts. This includes electric arc furnaces, recycled top gas from blast furnaces, and green hydrogen-based DRI. JSW is currently setting up a 3,800-tonne green hydrogen plant for steelmaking purposes and is working with international partners.

Acharya says the company’s work on steelmaking slag, which has traditionally been sent to landfill but is now processed into construction

JAYANT ACHARYA CEO, JSW STEEL



MANUFACTURING MINES AND METALS

→ JSW is doubling down on green technologies, renewable power, and circular practices, to grow sustainably

→ The company has set a target to reduce its CO₂ emission intensity by 42% by 2030, from a 2005 baseline

→ It is aiming for carbon

neutrality by 2050

→ JSW Steel has developed a sustainability framework covering 17 focus areas

→ JSW Steel has operationalised around 800 MW of renewable capacity, with plans to increase this to 2.5 GW

“We are scaling up the use of steel scrap in our furnaces to provide the dual benefit of reducing the energy consumption and carbon emissions, in line with our principle of material circularity and efficiency,” says Prabodha Acharya.

By 2030, the company aims to achieve an emission intensity of 1.95 tonnes of CO₂ per tonne of crude

sand and materials for cement production. JSW has implemented the direct injection of plastic waste into furnaces. Ferrous scrap processing has been improved using advanced shredder technologies, helping reduce reliance on virgin iron ore and cutting emissions. **BT**

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